National University of Computer & Emerging Sciences, Karachi
Fall-2017 CS-Department
Lab Final Exam

Course Code: EL-213	**Course Name:** Computer Organization & Assembly Lang Lab
Instructor Name(s): Syed Zain ul Hassan, Sehrish Saeed, Sumaiyah Zahid	
Student Roll No:	**Section No:**
Date:	**Time:**

Instructions:
Attempt all tasks
Return the question paper after exam

Max Marks: 20 points

Task1: (5 marks)

Write an assembly program to concatenate two strings taken from the user.

Sample Output:

Enter string 1: *FAST*
Enter string 2: *NUCES*
Concatenated string: *FASTNUCES*

Task 2: (5 marks)

Write a program which contains a procedure named *removeStr* that removes n characters from a string. Pass a pointer to the position in the string where the characters are to be removed. Pass an integer specifying the number of characters to remove.

Task 3 (a): (2.5 marks)

Write a program that contains a procedure to generate a random integer in the range given by the user. The procedure should be passed a lower bound m and an upper bound n.

Task 3 (b): (2.5 marks)

Write a program which contains a byte array of 20 elements. Use the procedure you made in *part a* to fill this array with integers in the range set by the user. The program should exchange each consecutive pair of integers in that array and then display its contents.

Task 4: (5 marks)

Write an assembly language function that receive two parameters the offset of string array and the string's size. It must return a count of the longest increasing sequence of vowel values.
For example, in the following string, the longest increasing sequence begins at index 3 and has a length of 3 {a,i,o}:

myStr BYTE "Coal is love", 0